

- ① A variable capacitor is connected to a 200V battery. If its capacitance is changed from $2\mu\text{F}$ to $x\mu\text{F}$, the decrease in energy of the capacitor is $2.0 \times 10^{-2}\text{J}$. Calculate the value of x .
- a) $1\mu\text{F}$ b) $2\mu\text{F}$ c) $3\mu\text{F}$ d) $4\mu\text{F}$
- ② Two parallel conductors carrying current of 4A and 10A are placed 2.5cm apart in vacuum. The force per unit length between them is :-
- a) $6.4 \times 10^5 \text{ N/m}$ b) $6.4 \times 10^{-2} \text{ N/m}$ c) $4.6 \times 10^{-4} \text{ N/m}$
 d) $3.2 \times 10^{-4} \text{ N/m}$
- ③ Use Bohr's model of hydrogen atom to obtain the relationship between the angular momentum and the magnetic moment of revolving electron.
- ④ An optical instrument uses a lens of power 100D ~~for~~ for objective lens and 50D for its eyepiece. When the tube length is kept at 25cm the final image is formed at infinity.
- a) Identify the optical instrument
 b) Calculate the magnification produced by the instrument.
- ⑤ a) State Gauss law for electrostatics and derive an expression for the electric field due to a long, straight and thin ~~conducting~~ uniformly charged wire (linear charge density λ) at a point lying at a distance r from the wire.
- (b) The magnitude of electric field (in NC^{-1}) in a region varies with the distance r (in m) as $E = 10r + 5$. By how much does the electric potential increase in moving from point $r = 1\text{m}$ to $r = 10\text{m}$